Please take

The Bonus of Course

Bonus book has

- 28 Topics
- 98 pages

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1 Introduction to Programming

Introduction to

- Program & Programming
- Programming Languages
- Python Programming Language
- Python IDEs

Program meaning

Program is also called

Software

Application

which is **created** by **programming**
for a machine to **perform** specified tasks

Examples for programs:

Google

facebook

Instagram

Programming meaning

Programming **is** the **process** or **activity** of **writing programs** for **computers** or **any kind of devices**

We write **programs** by **Programming Languages**

There are many kind of Programming Languages:

However we are going to learn Python Programming Language in this course

Python IDEs

What is **IDE**?

IDE is the short form of Integrated Development Environment.

IDEs are coding tools which are used to write codes.

The **most popular IDEs** are for **writing Python codes**:

IDLE

Jupyter Notebook

Spyder

Atom

Visual Studio Code

PyCharm

However in **this course**, we will use **Jupyter Notebook** and **PyCharm**

Anaconda Installation

If you want to download Anaconda for a different **OS** (e.g. **Mac** or **Linux**), go **the down** of download link, and you can choose Anaconda for a different **OS** as the bellow image:

My **OS** is **Windows**, therefore I choose Anaconda for Windows (64-Bit Graphical Installer).

Anaconda Download List:

Windows:

64-Bit Graphical Installer

32-Bit Graphical Installer

MacOS:

64-Bit Graphical Installer

64-Bit Command Line Installer

Linux:

64-Bit (x86) Installer (544 MB)

64-Bit (Power8 and Power9) Installer (285 MB)

Initializing Python

Write “python” to initialize **Python** on Anaconda Prompt:

Anaconda Prompt (anaconda3) - python

```
(base) C:\Users\sabr>python
Python 3.8.8 (default, Apr 13 2021, 15:08:03) [MSC v.1916 64 bit (AMD64
)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print('Hello World')
Hello World
>>> print('Hello Python, I can learn you:)')
Hello Python, I can learn you:)
>>>
```

3 Installing and Using Jupyter-Notebook

Jupyter-notebook: Installation

You can install jupyter by using **conda** or **pip. conda** and **pip** are used to installing **python packages (libraries)**

by **conda** →

```
conda install -c anaconda jupyter
```

by **pip** →

```
pip install jupyter
```

Note:

Jupyter-notebook should be installed automatically, when you **installed Anaconda**.
If it was not installed automatically, you can use above codes for installing jupyter-notebook

4 Print function

Print function: print(*object(s)*)

```
print(object(s))
```

object can be a **number**

Ex 1)

```
print(5)
```

```
5
```

objects can be **many numbers** (*sequence of numbers*). Separate each **number** with comma **,**

Ex 2)

```
print(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
```

```
0 1 2 3 4 5 6 7 8 9 10
```

objects can be *sequence of* **texts numbers.** Separate each item (text or number) with comma **,**

Ex 3)

```
print(0, 'Hello', 1, 'I', 'am', 2, 'am', 3, 'python', 4, 5, 6, 7, 8, 9, 10)
```

```
0 Hello 1 I am 2 am 3 python 4 5 6 7 8 9 10
```

5 Variables in Python

Variables in Python

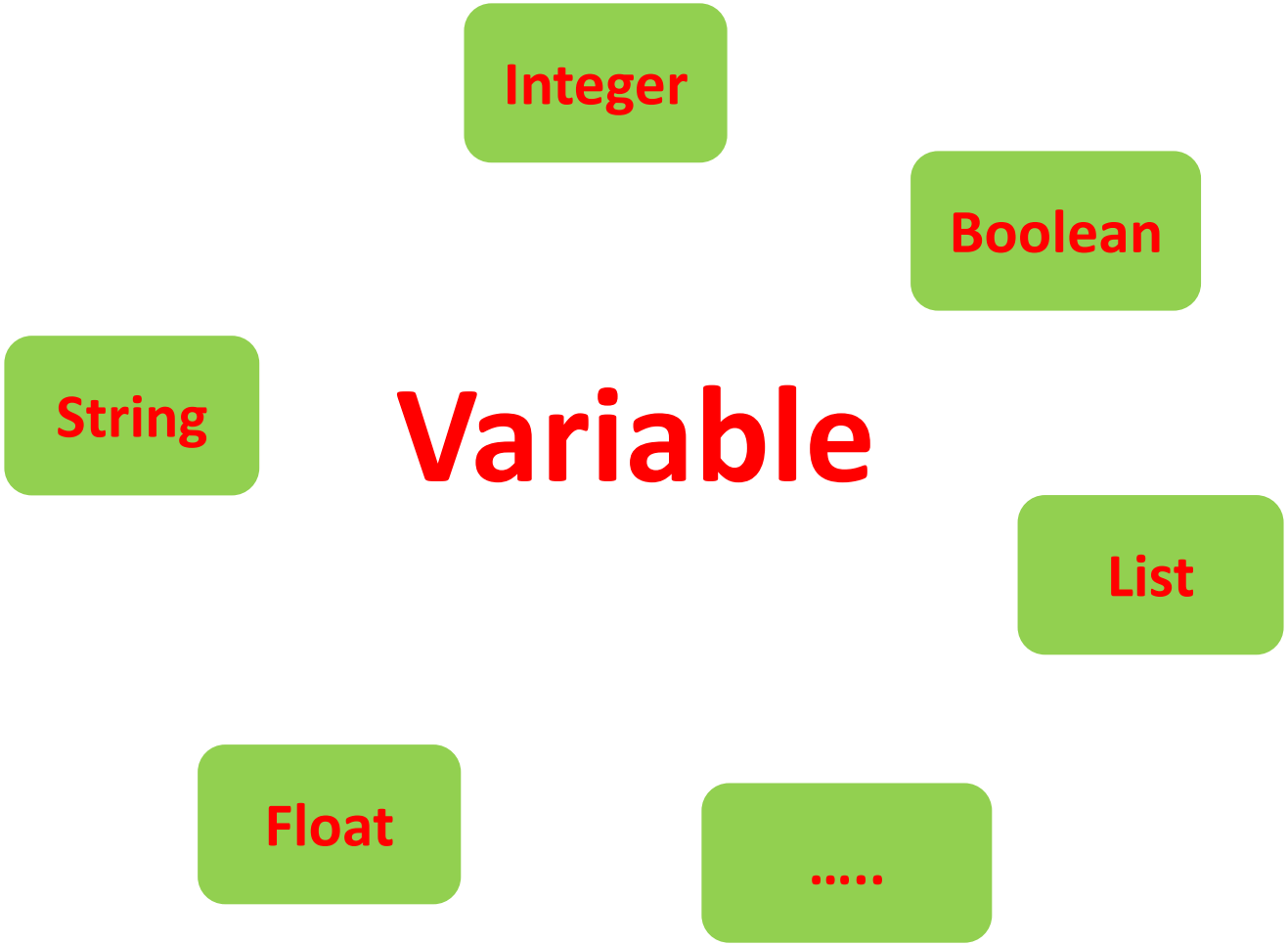
What is Variable?

Variable is a **container** for **storing data values**.

Variable =

Variables in Python

There are **many kind** of **Variables** in Python



Variables in Python: type() function

```
a = str(7)
```

Variable name

str() function

Integer value

type() function check the **type** of the Variable

```
print('data type',type(a))
```

```
data type <class 'str'>
```

Variable type (class) is String

Variables in Python

We have learnt

```
string_variable = 'Python'

integer_variable = 3

float_variable = 3.8
```

In future lessons,
We will learn

```
complex_variable = 3 + 2j

boolean_variable = True

list_variable = ['Python', 3, 3.8]

tuple_variable = (3, 'Python', 3.8)

set_variable = {3.8, 'Python', 3}

dictionary_variable = {'name':'Python', 'version':3}
```

Variables in Python

We have learnt

```
string_variable = 'Python'

integer_variable = 3

float_variable = 3.8
```

In future lessons,
We will learn

```
complex_variable = 3 + 2j

boolean_variable = True

list_variable = ['Python', 3, 3.8]

tuple_variable = (3, 'Python', 3.8)

set_variable = {3.8, 'Python', 3}

dictionary_variable = {'name':'Python', 'version':3}
```

6 Space Error and Tab Error

Space Error and Tab Error in Python

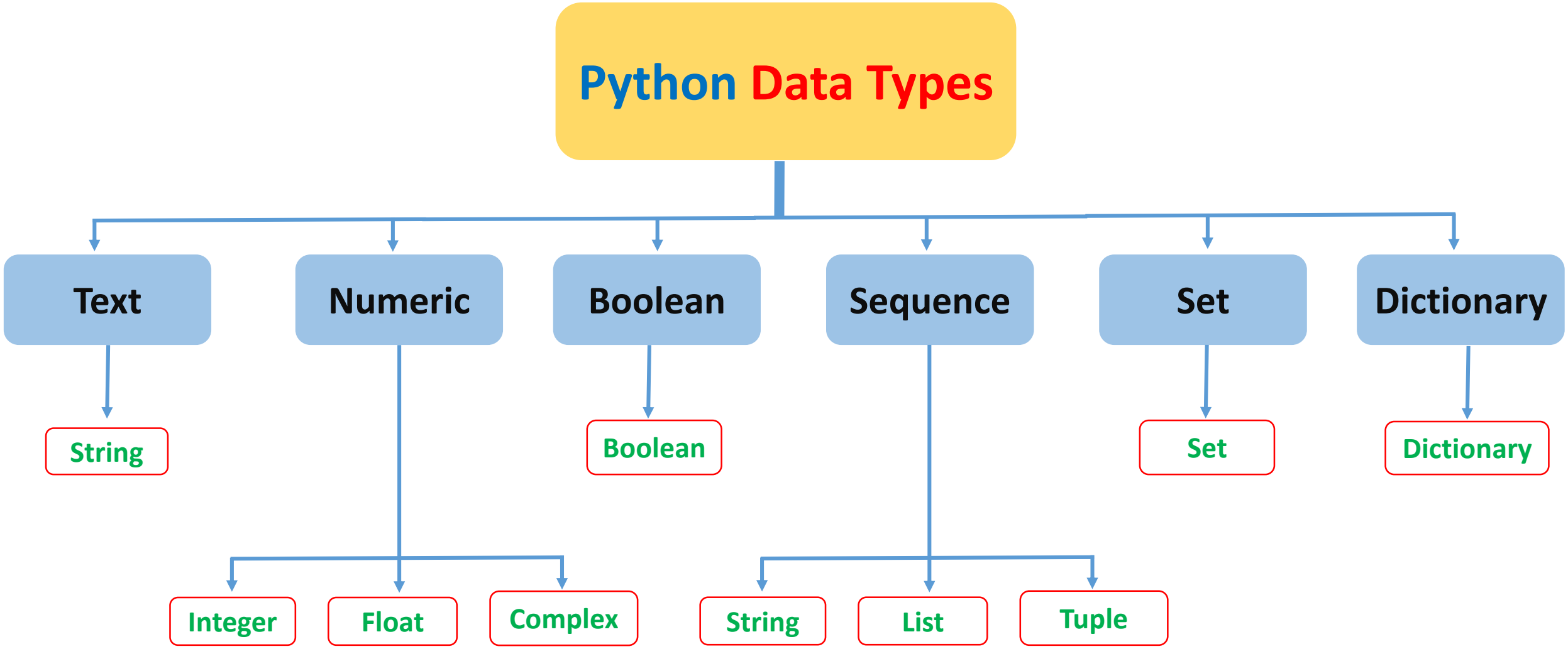
```
name = 'Jonny'
print(name)
```

```
Jonny
```

We have to use the same number of spaces or same number of tabs in the same block of code, otherwise Python will give you an **error**:

7 Python Data Types

Python Data Types: Introduction



Text Type: Single-line String

String Variable Value should be **in 'Single Quotation'** or **in "Double Quotation"** or **in str() function**

Single Quotations:

```
name_sq = 'Python'
print(name_sq)
```

```
Python
```

Single-line text

Double Quotations:

```
name_dq = "Python"
print(name_dq)
```

```
Python
```

Single-line text

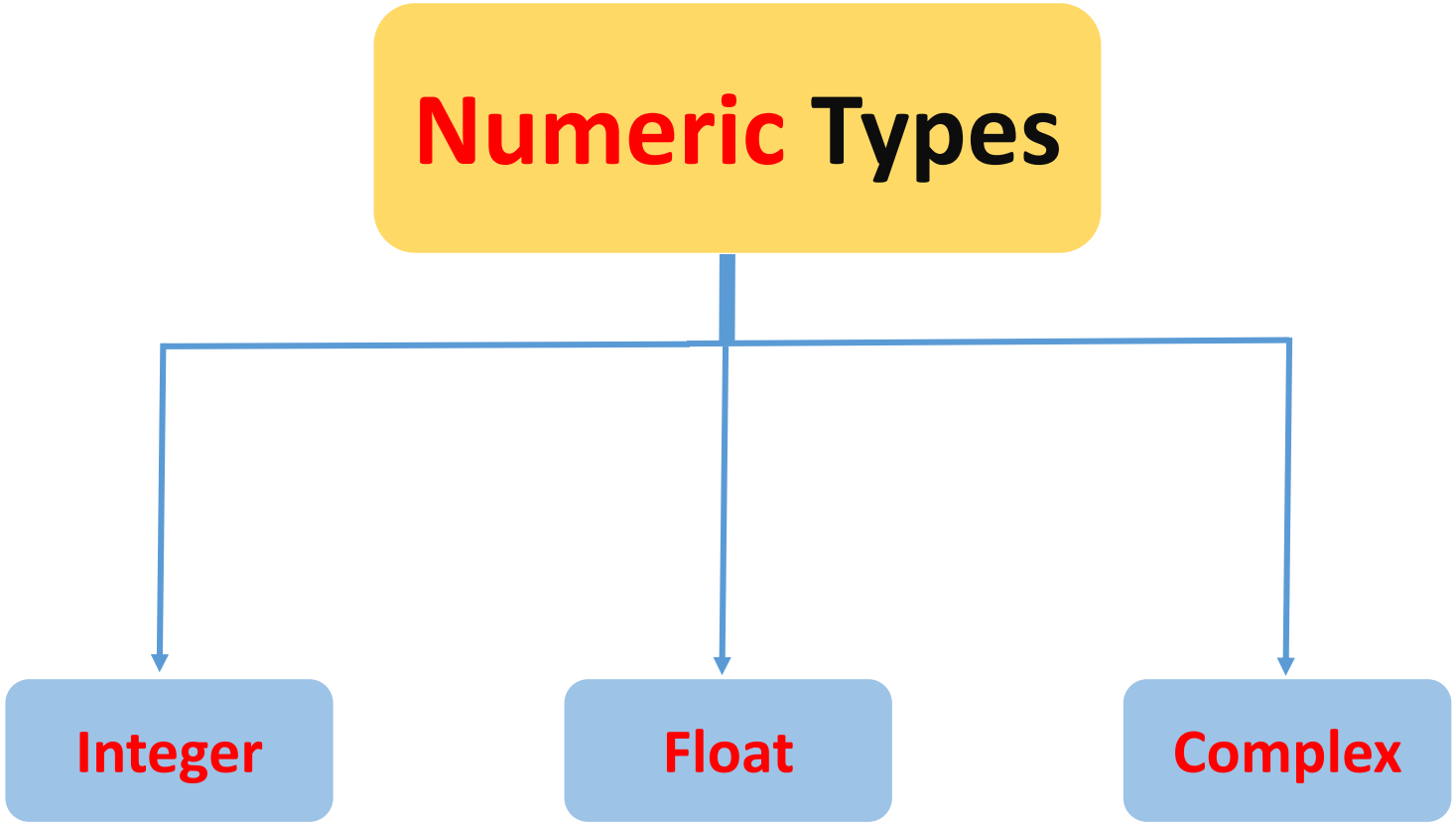
str() function:

```
age = str(25)
print(age)
```

```
25
```

Single-line text

Numeric Types: Integer, Float, and Complex



Numeric Types: Float

Or we can you use float() function for creating float **variables** value from string or integer **variables**.

But, if you want to **convert** string **variable to** float **variables:**

the **value** of string **variable must be number.**

Check Variable type by type() function

String **to** Float

```
age = '25'

age = float(age)
print(age)
```

```
25.0
```

```
print(type(age))
```

```
<class 'float'>
```

Integer **to** Float

```
number = 22

number = float(number)
print(number)
```

```
22.0
```

```
print(type(number))
```

```
<class 'float'>
```

Numeric Types: Complex

What is Complex number?

Complex number is similar to integer or float number, but it has the "j" after number at imaginary part.

Complex numbers are written with a "j" as the imaginary part:

Example:

Declaring complex variable

"j" character changes number to complex number

```
x = 1j
```

Complex variable name

number

Note: "j" must be written after number.
There is no any space between "j" and number

8 SyntaxError and NameError in Code

NameError in Code

If you **assign** undeclared (undefined) variable as the **value** of other **variable**, you will get **NameError** :

Ex 1) undefined (b) variable as the **value** of new **variable**

```
a = b

print(a)
```

```
---------------------------------
NameError
<ipython-input-1-ed99eac7bf8d> in <
----> 1 a = b
      2
      3 print(a)

NameError: name 'b' is not defined
```

Ex 2) undefined (b) variable as the **value** of new **variable**

```
b = a

print(b)
```

```
---------------------------------
NameError
<ipython-input-1-50082f0d24ed> in
----> 1 b = a
      2
      3 print(b)

NameError: name 'a' is not defined
```

9 Data Conversion

Data Conversion in Python

Integer
Float
Complex

to **String**

String
Integer
Complex

to **Float**

String
Float
Complex

to **Integer**

String
Integer
Float

to **Complex**

Data Conversion in Python: Convert to Float

Convert String, Integer, Complex **to** **Float by float() function**

You **can** convert **String** to Float by using float() function, if **string variable value** is **number** or **decimal number**.

String to Float

```
my_str = '10'

my_float = float(my_str)
print(my_float)
print(type(my_float))
```

```
10.0
<class 'float'>
```

String to Float

```
my_str = '10a'

my_float = float(my_str)
print(my_float)
print(type(my_float))
```

```
-----------------------
ValueError
```

Wrong conversion, because **string** value is not only number

10 Comments in Python

Comments in Python

Comments are used to **explain Python code**.

Comments are used to **make the code more readable**.

To make a comment, we can use bellow symbols:

1) Hash

```
# Text
```

Hash # is for a ***single-line comment***

2) Three single quotations

```
'''
Text
'''
```

3) Three double quotations

```
"""
Text
"""
```

Triple Quotes is for ***multi-line comments***

Note: **Hash #** is preferred for a ***single-line or for 2, or 3 line comments***. However, Triple Quotes is preferred for ***multi-line comments***

11 Booleans in Python

Booleans in Python

There are **2 types of Boolean values** in Python

True

```
my_bool = True

print(my_bool)
print(type(my_bool))
```

```
True
<class 'bool'>
```

False

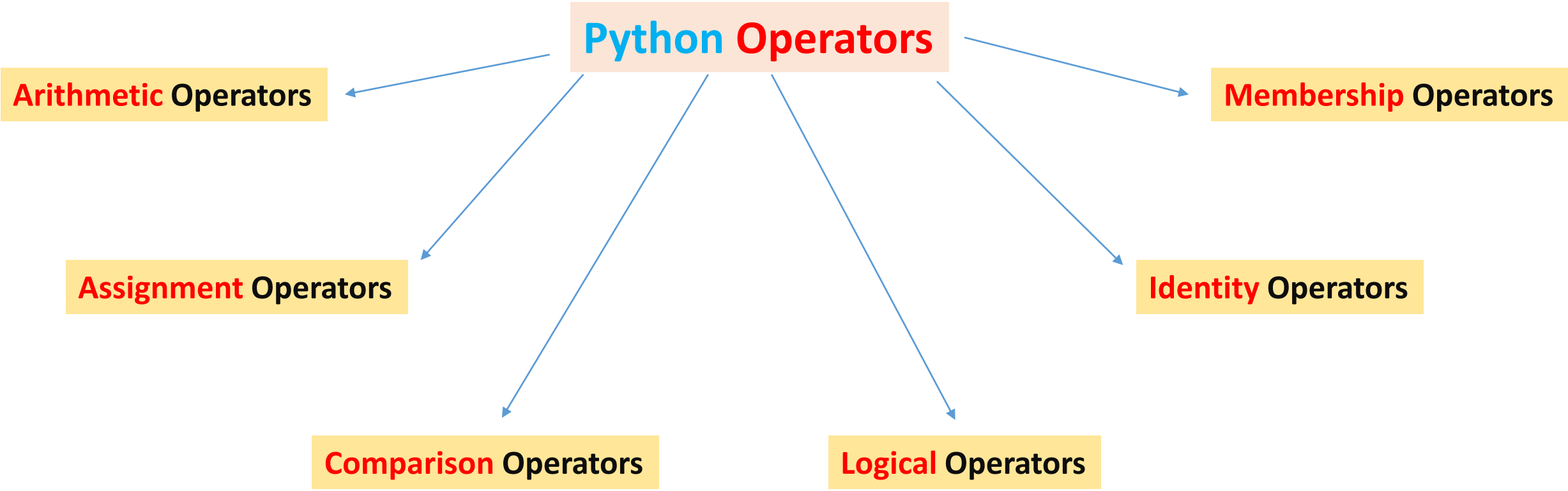
```
my_bool = False

print(my_bool)
print(type(my_bool))
```

```
False
<class 'bool'>
```

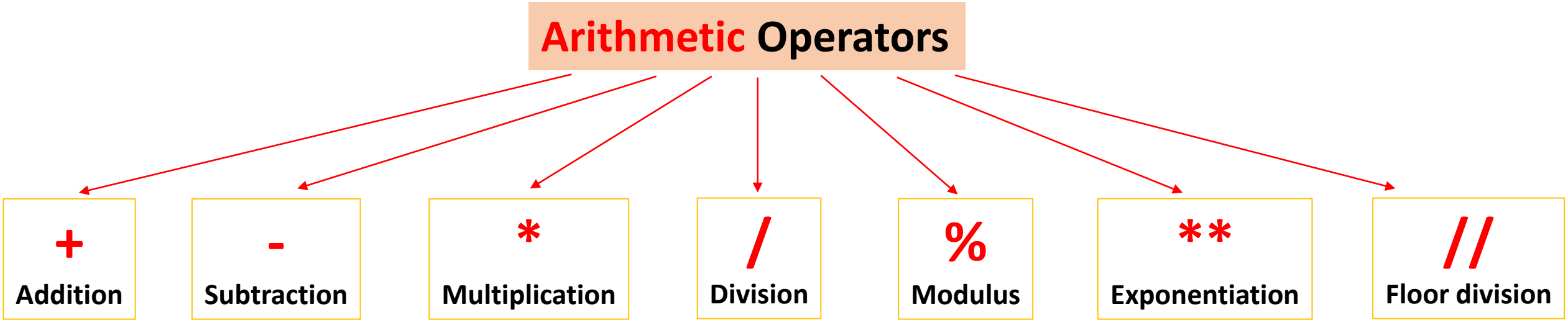
12 Python Operators

Python Operators



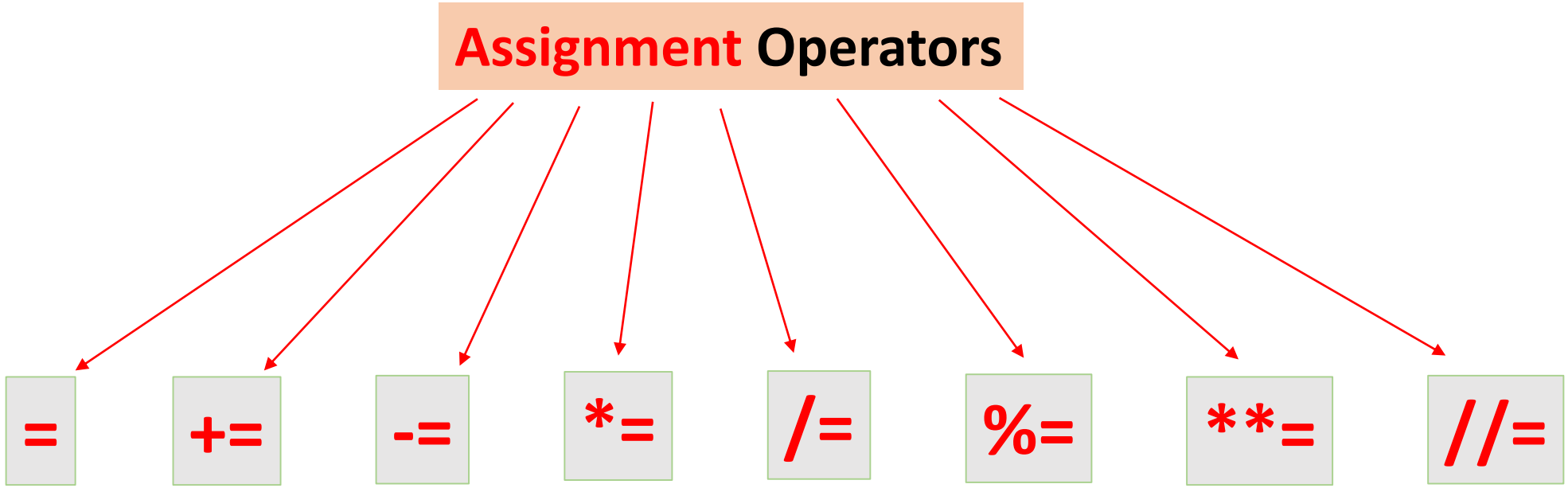
Operators can be used to perform operations on variables and values.

Python Operators: Arithmetic Operators



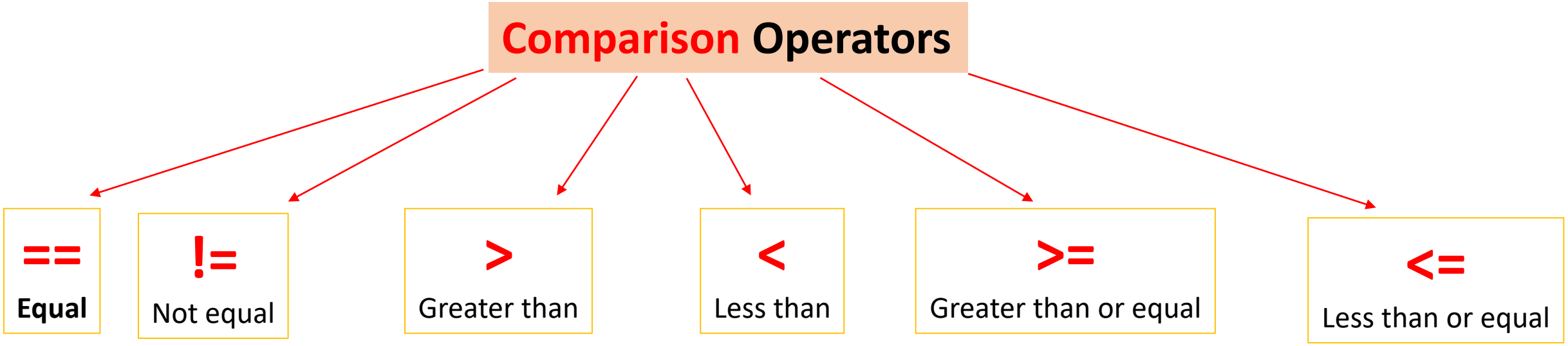
We use **Arithmetic operators** with **numeric values** to **perform** common **mathematical operations**.

Python Operators: Assignment Operators



Assignment operators is used to ***assign values*** ***to variables***

Python Operators: Comparison Operators



We use **Comparison operators** to **compare** two values or variables.

Note: The result of Comparison will be equal to **Boolean value** (True or False)

Python Operators: Logical Operators

Logical Operators

- **and** → Returns **True** if all condition statements are true
- **or** → Returns **True** if one of all condition statements is true
- **not** → **Reverse the result**, returns **False** if the result is true

We use **Logical operators** to **combine two** or **many** **conditional statements.**

Note: The result of Logical operators will be equal to **Boolean value** (**True** or **False**)

Python Operators: Identity Operators

Identity Operators

- **is** → Returns **True** if both variables are the same object, else returns **False**
- **is not** → Returns **True** if both variables are not the same object, else returns **False**

Identity operators are used to compare the **variables**, not if they are equal, but if they are actually the same object, with the **same memory** location.

Note: The result of Identity operators will be equal to **Boolean value** (**True** or **False**)

Python Operators: Membership Operators

Membership Operators

- **in** → Returns **True** if a sequence with the **specified value** is present in the object, else returns **False**
- **not in** → Returns **True** if a sequence with the **specified value** is not present in the object, else returns **False**

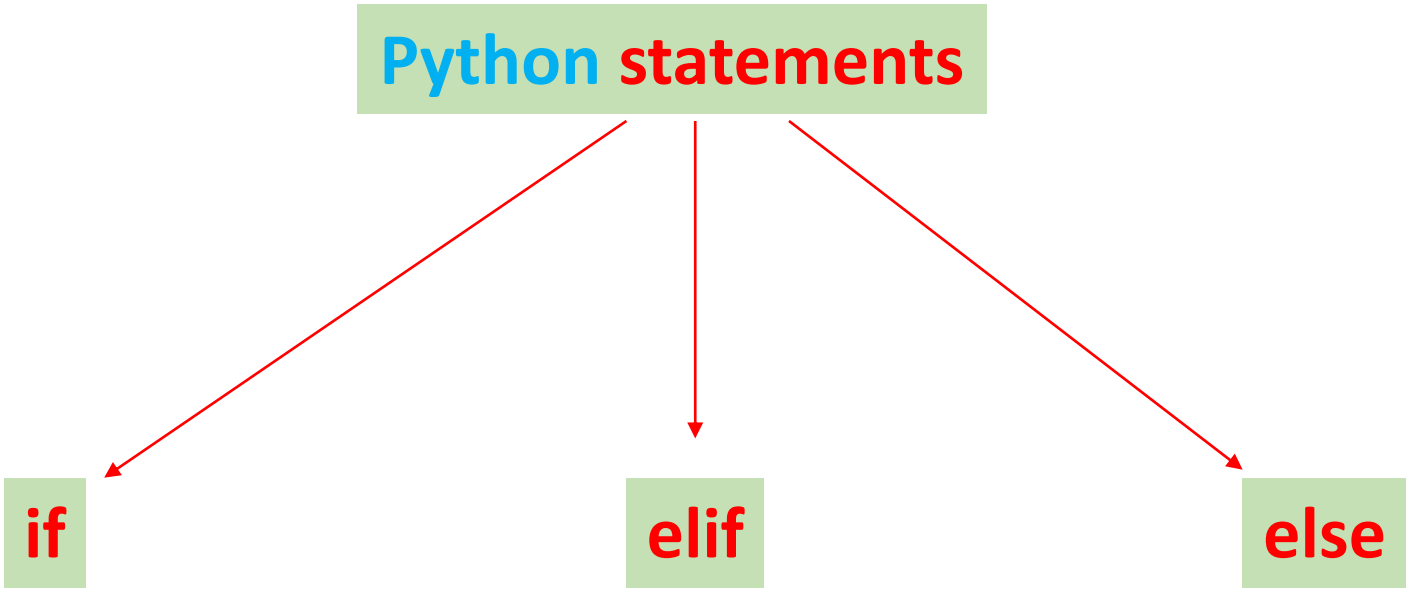
Membership operators are used to **check** if a **sequence** is **presented** in an **object**

.

Note: The result of Membership operators will be equal to **Boolean value** (**True** or **False**)

13 Python Statements

Python statements



If statement

If Boolean value is **False**, the code inside of "**if statement**" doesn't **work**.

Comparing two values. If Comparison is **False**, the **"Correct" will not be printed:**

```
if 10 < 9:
    print("Correct")
```

if statement

```
if 10 < 9:
    print("Correct")
```

Whitespace: **a Tab** (Indentation)

If statement: pass

if statement cannot be empty, however if you don't want this statement to return to any output, just put pass **inside** if statement.
pass statement passes if statement, and it avoids getting error.

Comparison is True

```
if 10 > 5:
    pass
```

pass statement passes if statement.
it avoids getting error.

No output, because
of pass statement

14 Python Loops

Python Loops

There are **two** types of Loops in **Python**:

while

for

For Loop: The break statement

The **break** statement can **stop** "**the for** loop" before "the **for** loop" **looped** through all the items

```
my_numbers = "123456789"

for n in my_numbers:
    print(n)
    if n == '5':
        break
```

break stopped the loop if n == '5'

```
1
2
3
4
5
```

The output is printed until n==5.

For loop: range() - specific increment value

The range() function the default increment value is **1**, however it is possible to change the increment value by adding a third parameter: range(0, 10, **2**):

```
for a in range(0, 10, 2):
  print(a)
```

1st parameter | 2nd parameter | 3rd parameter

outputs

```
0
2
4
6
8
```

You used *2 as* **the specific increment value** on the 3rd parameter of range() function, therefore outputs are 0, 2, 4, 6, 8

15 Python Arrays

Arrays

What is an Array?

If the variable **holds more than one value at a time,** it is called the Array.

The groups of Arrays in Python: **String, List, Tuple, Set, Dictionary**

Sequence types

Set type

Dictionary type

```
str_array = 'Python'

list_array = [1, 2, 3, 4]

tuple_array = (1, 2, 3, 4)

set_array = {1, 2, 3, 4}

dictionary_array = {'one':1, 'two':2, 'three':3, 'four':4}
```

Note: Python does not have built-in support for Arrays, but **Python Lists** can be used instead Arrays

We will learn all of them one by one.

16 Python Strings

Python Strings: Index Order

Note: **String element** starts **from 0** ***not 1***

1st = 0
2nd = 1
3rd = 2
...

```
str_variable = 'Python'
```

Index Order

0	1	2	3	4	5
1st	2nd	3rd	4th	5th	6th

If you want to print **"P"** from string variable:

```
print(str_variable[0])
```

```
P
```

If you want to print **"h"** from string variable:

```
print(str_variable[3])
```

```
h
```

17 Python Lists

Python Lists

Lists can store **multi-different types of values** in a single variable.

Integer Float String Boolean

```
list_variable = [1, 2.0, "three", True]

print(list_variable)
```

```
[1, 2.0, 'three', True]
```

Python Lists: Items, Negative Indexing

Negative indexing is to **start** from the end.

-1 is **1st last** item of **List**, -2 is **2nd last** item of **List** and so on

```
list_variable = [1, 2.0, "three", True]
```

-4 -3 -2 -1

Print **1st last** item of **list_variable**:

```
list_variable = [1, 2.0, "three", True]

print(list_variable[-1])
```

```
True
```

Python Lists: Looping Using List Comprehension

List Comprehension is the shortest syntax to loop through lists.

```
list_variable = ['Alex', 'John', 'Anna']
[print(a) for a in list_variable]
```

Everything in one line

```
Alex
John
Anna
```

With **list comprehension**
you can do all **in one line of code**

```
[print(a) for a in list_variable]
```

=

Without list comprehension
you must write 2 line code

```
for a in list_variable:
    print(a)
```

18 Python Tuples

Python Tuples: Items, Index order

Tuple **items** are **indexed**, **1st** item is index [0], **2nd** item is index [1] and so on

```
tuple_variable = (1, 2.0, "three", True)
```

0 1 2 3

Print **1st** item of **tuple_variable**:

```
tuple_variable = (1, 2.0, "three", True)

print(tuple_variable[0])
```

```
1
```

Print **3rd** item of **tuple_variable** :

```
tuple_variable = (1, 2.0, "three", True)

print(tuple_variable[2])
```

```
three
```

Python Lists: len() function

To know **how many items** a **tuple** has, use the **len()** function.
len() function shows **number of items of your tuple**

The **name** of Tuple variable

```
tuple_variable = (1, 2.0, "three", True)

print(len(tuple_variable))
```

Total values: 4

19 Python Sets

Python Sets

Set is a **Set type data**.

Set is a type of Python Arrays.

Sets are used to **store multiple items** in a single variable.

Set is a **collection** which is ***unordered***, ***unchangeable***, and ***unindexed,*** do ***not allow duplicate values***.

Sets are created using **curly brackets {}** or using **set()** function

using **curly brackets {}** to create **Set**

```
set_variable = {1, 2, 3, 4, 5}

print(set_variable)
```

```
{1, 2, 3, 4, 5}
```

using **set()** function to create Set

```
set_variable = set((1, 2, 3, 4, 5))

print(set_variable)
```

```
{1, 2, 3, 4, 5}
```

Python Sets: len() function

To know **how many items** a Set has, use the **len()** function.
len() function shows **number of items of your Set**

The **name** of Set variable

```
set_variable = {1, 2.0, "three"}

print(len(set_variable))
```

Total values: 3

Python Sets: Join Two Sets, update()

update() method **adds items** from another set into the current set

```
set_1 = {'A', 'B', 'C'}
set_2 = {'D', 'E', 'F'}

# adding items of set_2 into set_1

set_1.update(set_2)
print(set_1)
```

```
{'A', 'E', 'C', 'D', 'F', 'B'}
```

20 Python Dictionary

Python Dictionaries: key:value

Dictionary item is **divided two parts**: 'key' : 'value'

'key' is **key name** of Dictionary Item

Dictionary = {'key' : 'value'}

'value' is the value of Dictionary Item

Python Dictionaries : for loop, search values

If you want to search **values of items**, use **keys** of items as the **index** of **Dictionary:**

```
dictionary_variable = {'name':'Alex', 'age':25}

for d in dictionary_variable:
    print(dictionary_variable[d])
```

```
Alex
25
```

Note: You can use **keys** as index of **Dictionary**, but you ***cannot*** use 0, 1 … etc as index of **Dictionary**

21 Python Functions

Python Functions: Call function

You can **use function**, *how many times* you want by calling **function name**.

If you call **function 1** *time*,
it runs **1** *time*

```
def function():
    print('This is my function')


function()
```

```
This is my function
```

If you call **function 2** *times*,
it runs **2** *times*

```
def function():
    print('This is my function')

function()
function()
```

```
This is my function
This is my function
```

If you call **function 5** *times*,
it runs **5** *times*

```
def function():
    print('This is my function')

function()
function()
function()
function()
function()
```

```
This is my function
This is my function
This is my function
This is my function
This is my function
```

Python Functions: Parameters

Function with **no parameters :**

```
def info()
```

no **parameters**, **empty**

You can write **one** or **many parameters** in the **brackets** after **function name**.

Function with **parameters:**

```
def info(name, age, gender)
```

parameter 1 **parameter** 2 **parameter** 3

comma **comma**

If you write **2** or **more than 1 parameter** in the **brackets** , you have to write **comma ,** after **each parameter**

Python Functions: Parameters and Arguments

Function with **1 parameters** and 1 **argument**:

```
def function(name):

    print(name)

function('Alex')
```

Giving **value** to **parameter**

```
Alex
```

Function with **5 parameters** and 5 **argument**:

```
def function(fname, lname, age, gender, score):

    print(fname)
    print(lname)
    print(age)
    print(gender)
    print(score)

function('Alex', 'Jonh', 25, 'Male', 95.5)
```

Giving **values** to **parameters**

```
Alex
Jonh
25
Male
95.5
```

Note: ***Arguments*** are often shortened to ***args*** in **Python *documentations***

Python Functions: Parameters' Order

When you are declaring **Parameters** in function, first you must write **non default Parameters**, **after that** you write default Parameters

```
def student(age, name = 'Alex')
```

non default Parameters will be written before default Parameters

Python Functions: if, elif, else statements

You can use **if, elif, else** statements in your function:

```
def function(number):

    if number == 0:
        print('number is 0')
    elif number == 1:
        print('number is 1')
    else:
        print('number is not 0 or 1')

function(0)
function(1)
function(2)
function(3)
function(4)
```

```
number is 0
number is 1
number is not 0 or 1
number is not 0 or 1
number is not 0 or 1
```

22 Lambda Function

Python Lambda

Lambda is a function.
Lambda function is a small **unnamed** (***anonymous)*** function.

You can declare **any number of arguments** after **lambda function**

But, Lambda function can have **only one expression**

lambda arguments : **expression**

colon must be written after arguments

Python Lambda: Use lambda inside other function

Examples:

Ex 1)

```
def def_function(a):

    return lambda b: b * a

print(def_function(a = 4)(b = 10))
```

```
40
```

Ex 2)

```
def def_function(a):

    return lambda b: b + a

print(def_function(a = 50)(b = 30))
```

```
80
```

23 Input Function

Python input() function

There is **input() function** in Python, which is used to for user input.

When you use **input() function**, it asks input from user

name variable

```
name = input()
```

If your variable is **equal** to **input() function**, after *running your code*, **input() function** asks you to input **value** for **name** variable

input **value** for **name** variable

```
name = input()
Alex
```

After entering input **value**, please **Enter** button

```
name = input()
```

Output of **name** variable

```
Alex
```

24 Python Classes and Objects

Python Classes and Objects: Create Class

Python is an object oriented programming language.

Object-oriented Programming (OOPs) is a **programming paradigm** (***example***) that uses objects and classes in **programming**.

What is a Class?

Class is the blueprint which is used to create one object or many objects

We can **create** a **class** by using **class keyword**:

Class name

```
class myClass:

    number = 10
```

Colon after **Class name**

number is a **property** of **myClass**, and **it** is equal to **10**

Python Classes and Objects: Modify Object Properties

If you want, you can **modify** Properties on **objects**:

before **modifying**, name='Anna', age=30

```
class Student:

    def __init__(self, name, age):
        self.name = name
        self.age = age

student = Student('Anna', 30)
print(student.name, student.age)

#modifying properties by object
print('----------')

student.name = 'Alex'
student.age = 25
print(student.name, student.age)
```

modify Properties

before **modifying**

```
Anna 30
----------
Alex 25
```

after **modifying**, name='Alex', age=25

Python Classes and Objects: Polymorphism

What is Polymorphism?

Polymorphism is taken from the Greek **words**:

Poly = many
morphism = forms

Polymorphism = many forms

It means that we can **use** the **same function name** for **different targets**.

Example: Cell phone can be Polymorphism, you can **use** it for **many different targets, task, and so on**

Cell phone is Polymorphism

Cell phone can be used

- to take picture
- to talk
- to watch video
- to study

25 Python Inheritance

Python Inheritance: Introduction

What is **Inheritance**?

Inheritance is the ability of **one class** to **derive (inherit)** all methods and properties from **another class**.

There are **two type of classes** in **Python Inheritance**

Parent class

Child class

Parent class is the **class being inherited** from. **Parent class is** also called **Base class** or **Superclass**.

Child class is the **class** that ***inherits from another class***. **Child class is** also called **Derived Class, or Subclass**.

26 Python Scopes

Python Scope: Enclosing Scope, Enclosing Variable

Enclosing Scope is almost similar to **Local** Scope. But, **Local** Scope will be inside of **Enclosing** Scope :

```
def outer():

    name = 'Alex'

    def inner():

        age = 25

        print(name, age)

    inner()

    print(name)
    #print(age)

outer()
```

```
Alex 25
Alex
```

name variable is **local** variable for **outer()** function
But, **it** is **enclosing** variable for **inner()** function. Therefore, you can use it inside **inner()** function

age variable is **local** variable for **inner()** function

Blue rectangle is **Local** scope and it is the **region** to **use** age variable

Red rectangle **is** **Enclosing scope** and **it** is the **region** to **use** name variable.

Python Scope: Global Keyword

global keyword is used to **convert** local variable or enclosing variable to **global** variable

```
def student():
    global name
    name = 'Alex'
print(name)
```

```
Alex
```

global keyword makes name variable to be global variable

name variable can be used in anywhere, because **it** is **global variable**

27 Python Error Handling

Python Try Except: Many Exceptions

If you want, you can use **multiple** except **blocks** for handling error.

Also you can use the **name** of **occurred** error with except **block:**

```
print(name)

---------------------------------------
NameError
<ipython-input-1-9ba126b17b03> in <mod
----> 1 print(name)

NameError: name 'name' is not defined
```

Using **NameError** with except **block**

1st except **block**

2nd except **block**

```
try:
    print(age)

except NameError:
    print('age is not declared')

except:
    print('Error is occured, please check your code')
```

1st except **block** handled error, therefore **output** is → `age is not declared`

28 Python Modules

Python Modules

What is a **Module**?

Module is the **code library** of Python

Module is just **.py** (python) **file**.

Module can be **used** through **other .py** (python) **file**

There are **3 kind** of **Modules in Python**

Your **own Modules**

Your own python **libraries,** which are created by yourself

Built-in Modules
(Internal **Modules**)

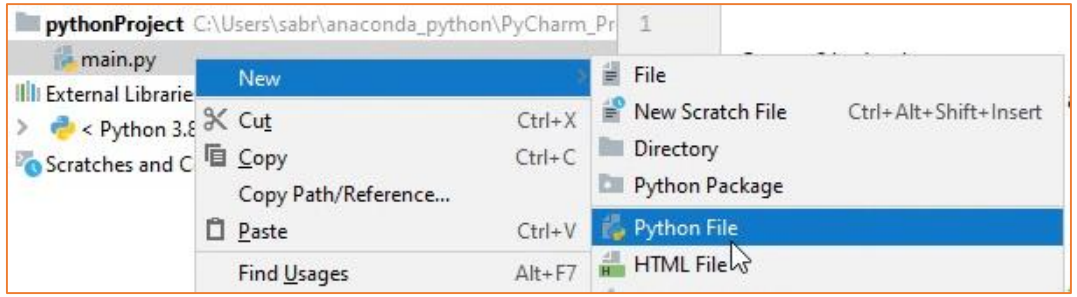
existed python **libraries,** and they are **already installed** when you **installed python**

External Modules

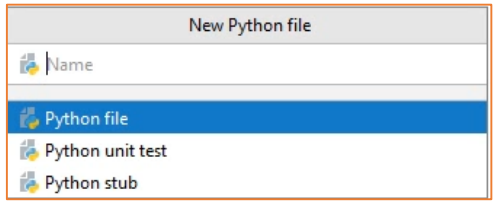
existed python **libraries,** But you have to **install** them **to use**.

Python Modules: Create your own Module

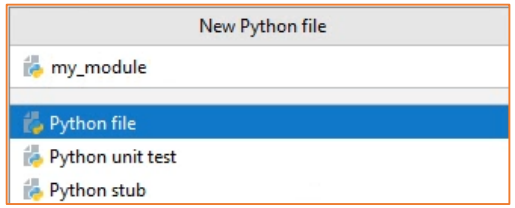
1. Create new **.py** file. Go **pythonProject** or main.py and crick **right side** of your **mouse** on **pythonProject** or main.py and **New** and click **Python File**



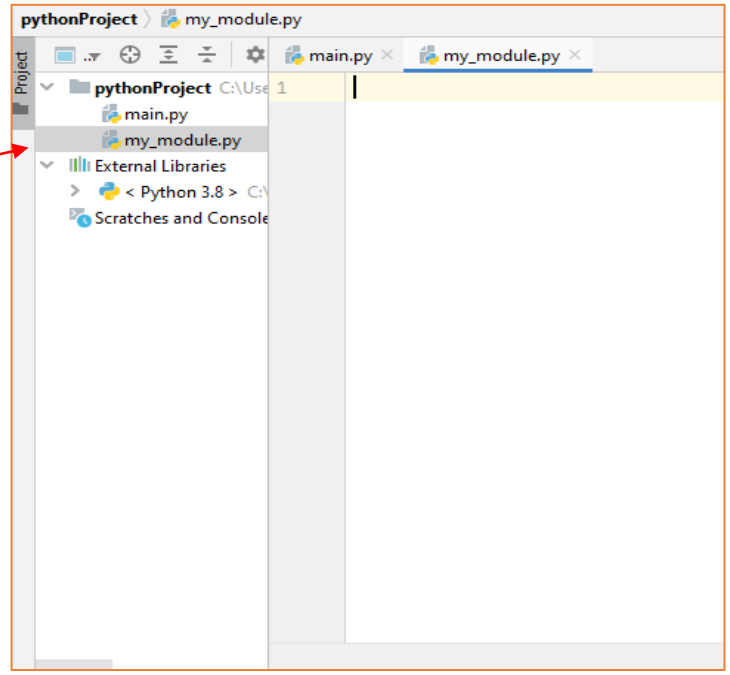
2. Give **name** to Python file



3. **name** is **my_module. Choose Python File**



4. my_module.py file is **created,**
That means **you created your own module.**
You can **write** your **codes** inside it



Python Modules: Renaming Module

import keyword used to **load** or **call** existed modules.

When you are **importing a module**, you can rename it by using **as keyword**

```
#rename module
import my_module as mm

student = mm.Student('Alex', 25, 'Male')
student.show_info()
```

After running, output:

```
Student name =  Alex
Student age =  25
Student gender =  Male
```

```
#rename module
import my_module as own_module

student = own_module.Student('Alex', 25, 'Male')
student.show_info()
```

After running, output:

```
Student name =  Alex
Student age =  25
Student gender =  Male
```

```
#rename module
import my_module as person

student = person.Student('Alex', 25, 'Male')
student.show_info()
```

After running, output:

```
Student name =  Alex
Student age =  25
Student gender =  Male
```

Python Modules: External Modules

What are External Modules?

External Modules are also the **code library** of **Python**, but you **have to install them** to **use**.

You can install **External Modules**, by using **their package names**

Module name	Package name	Import Module *after installation*	Installing by pip or conda
numpy	numpy	import numpy	pip install numpy or conda install numpy
cv2	opencv-python	import cv2	pip install opencv-python
pandas	pandas	import pandas	pip install pandas or conda install pandas

Python Modules: Installing External Modules

For example, you want to install numpy **package**, for *installing it* you can use:

Anaconda Prompt

using pip:

```
Anaconda Prompt (anaconda3)
(base) C:\Users\sabr>pip install numpy
```

using conda:

```
Select Anaconda Prompt (anaconda3)
(base) C:\Users\sabr>conda install numpy
```

Pycharm Terminal

using pip:

```
> pip install numpy
TODO  Problems  Terminal
```

using conda:

```
> conda install numpy
TODO  Problems  Terminal
```

Jupyter Notebook

using pip, type ! before pip

```
!pip install numpy
```

using conda, type ! before conda

```
!conda install numpy
```